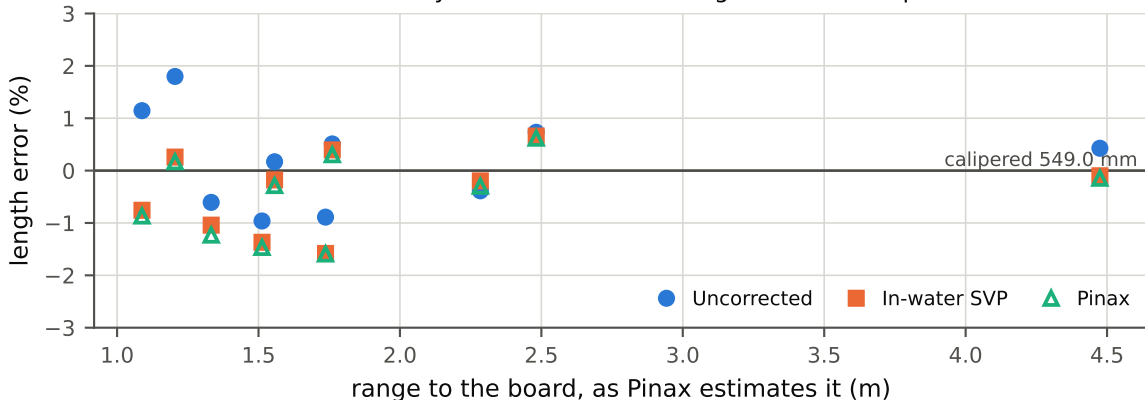


Camera-only end to end: measuring a 549.0 mm span



Ten pool frames; each model calibrates the laser with itself, so every pipeline is internally consistent. Median +0.3 / -0.2 / -0.3 %, worst 1.8 / 1.6 / 1.6 %, in legend order. All three land on the calipered span, including the uncorrected one: its range and magnification errors are both about a quarter, and they cancel. Length cannot separate these models — §4.1's ground-truth-free test is what does.